Wind and Solar Radiation in Switzerland

A brief description of measurements

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1 Data and methods

1.1 Source and variables

I use Meteoswiss open data at daily and montlhy level. I am mainly interested in wind speed and solar radiation: electricity production sources for wind turbines and solar panels. From Meteoswiss database, I rely on mean wind speed in m/s (at daily and monthly levels: `fkl010d0`, `fkl010m0`) and on mean solar radiation in W/m^2 (at daily and monthly levels: `gre000d0`, `gre000m0`).

I retains starting values for wind and radiation measurement from 1980 for wind and 1981 for solar radiation[1]. Thus I am able to perfom an analysis from 1980/1981 to 2025[2]. Nevertheless, some stations across Switzerland and across time do not measure either mean wind speed or mean solar radiation, or both. That is why, for a better consistency I choose to use 2 stable datasets: a first from 1980 to 2025, as second from 2015 to 2025. On the 1980–2025 period, only 42 stations measure both wind and solar variables from 1980 to 2025. On the 2015–2025 period, 124 stations measure wind and radiation. For a better identification, I indicate on each graphs and maps the number of stations used in the sample. A station is called *stable* if it has at least one non-missing daily measurement in every year of the selected period. This does not require complete daily or monthly coverage.

Despite a first cleaning done by Meteoswiss, outliers still persists. Thus, I retain values lying within 0–40 m/s for daily mean wind speed, 0–30 m/s for monthly mean wind speed, 0–500 W/m^2 for mean daily solar radiation and 0–400 W/m^2 for monthly mean solar radiation (inclusive bounds). This bounds correspond to anormal measurements at stations level.

Period in saved output	Wind-stable	Solar-stable	Stable for both
1980–2025 (solar: 1981–2025)	42	42	42
2015–2024	145	129	123

Table 1: Counts of observartion across sub datasets.

[1] First radiations measures started in 1981, nothing before

[2] I wil extend to 2026 I guess

2 Monthly seasonal profiles

2.1 Overall seasonal cycle

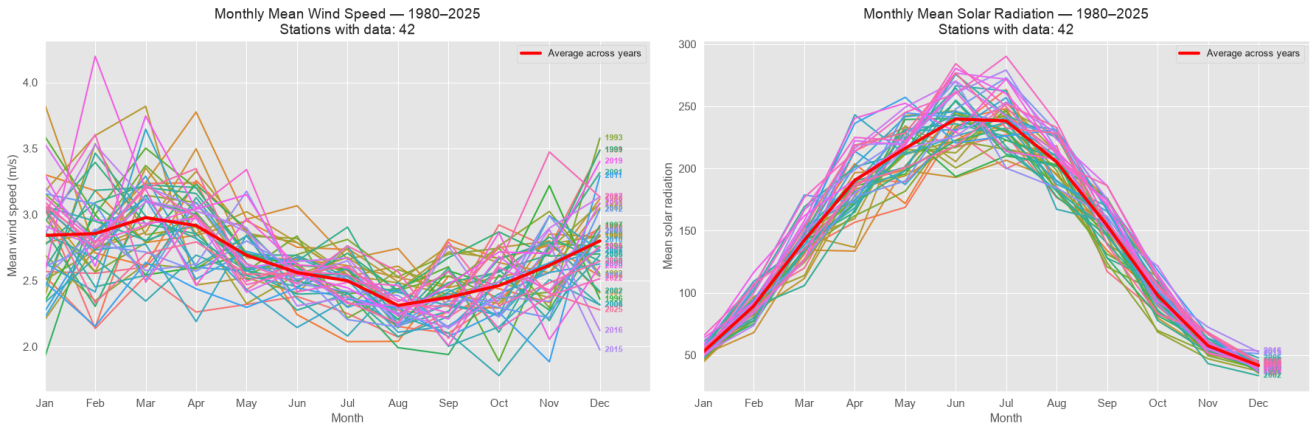


Figure 1: Overall monthly profiles, 1980–2025.

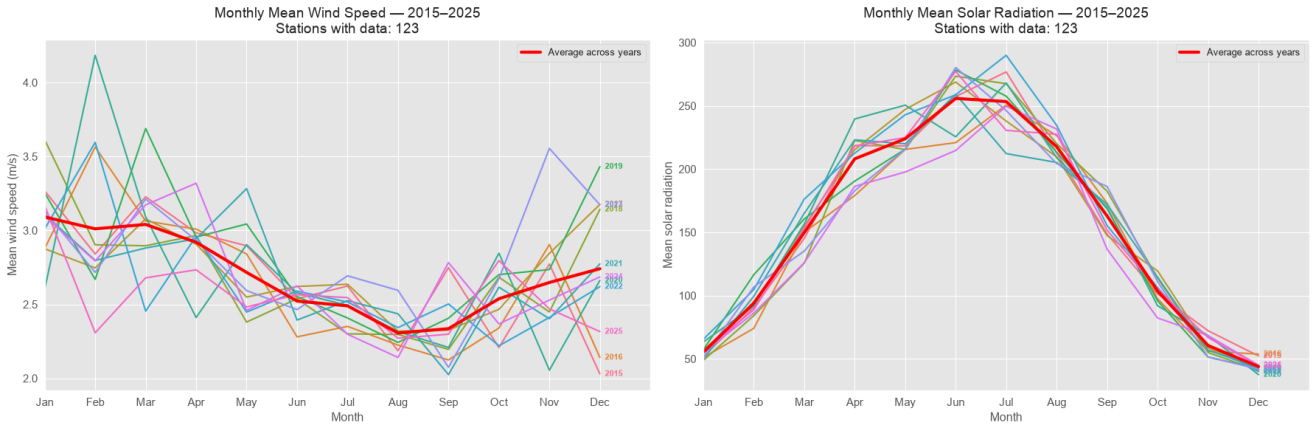


Figure 2: Overall monthly profiles, 2015–2025.

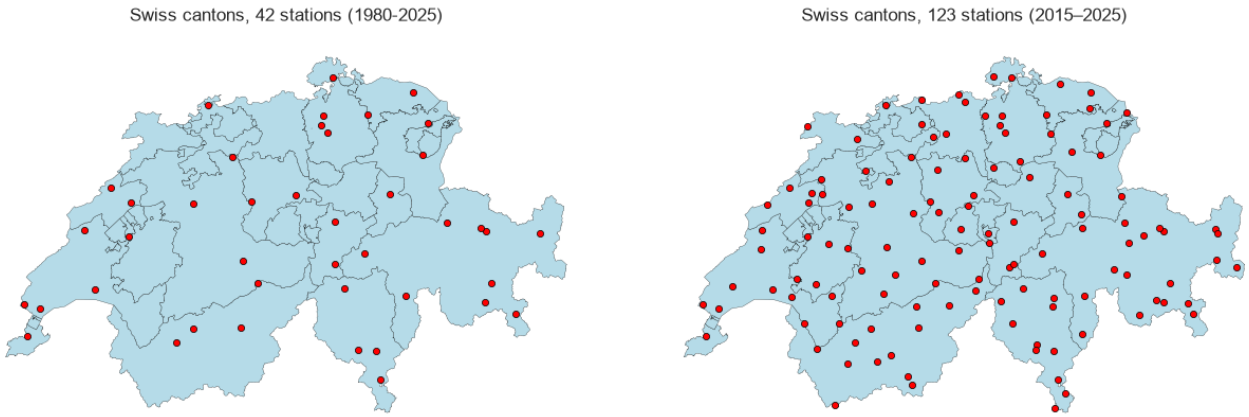


Figure 3: Locations of the stable stations: 42 stations for 1980–2025 (left) and 123 stations for 2015–2025 (right).

2.2 Differences by elevation

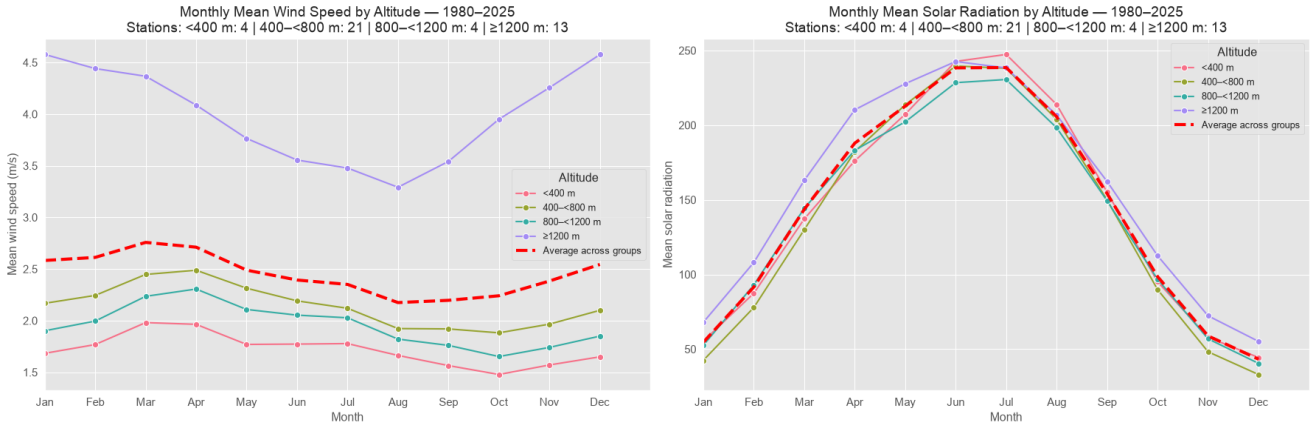


Figure 4: Monthly profiles by elevation, 1980–2025.

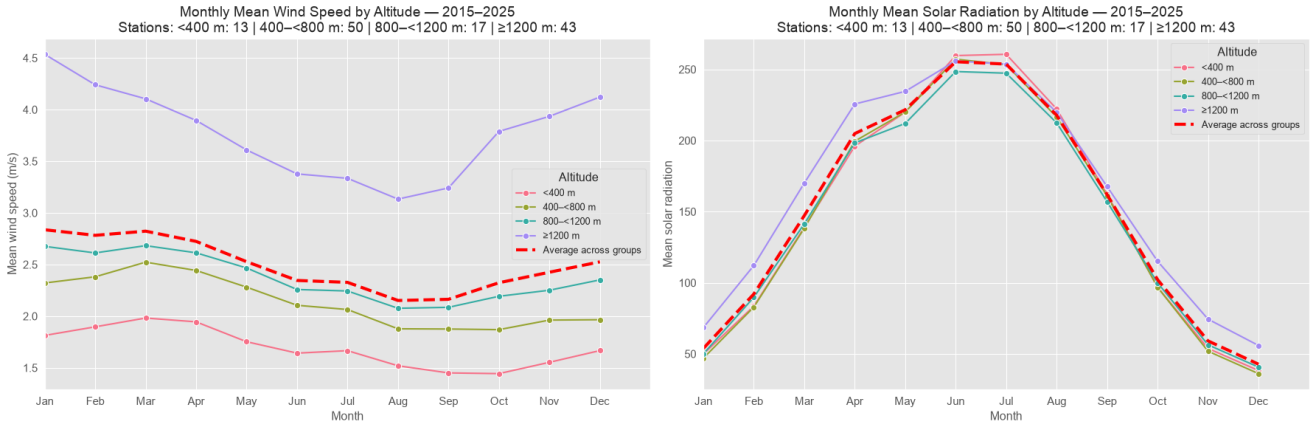


Figure 5: Monthly profiles by elevation, 2015–2025.

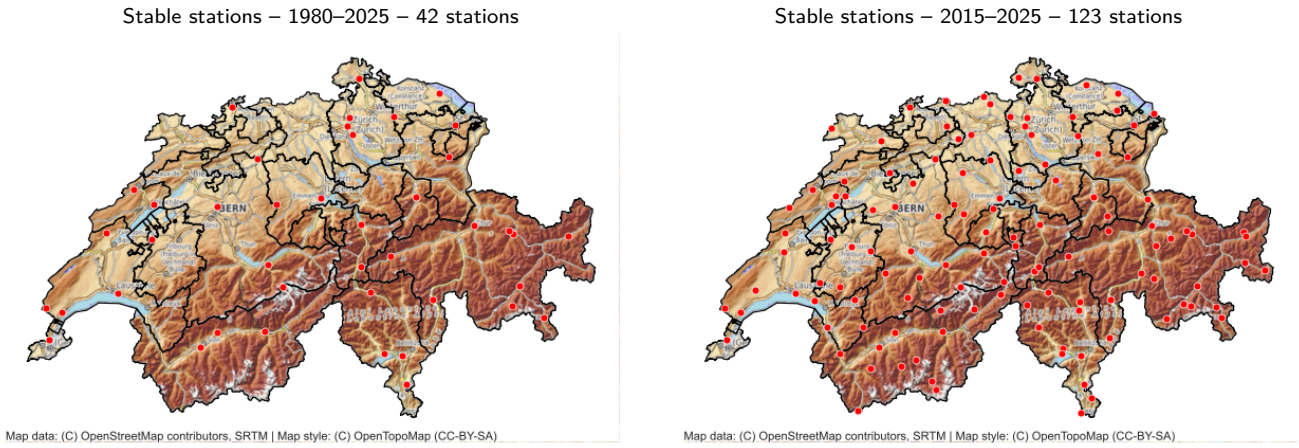


Figure 6: Station locations over the Swiss terrain: 1980–2025 (left) and 2015–2025 (right).

2.3 North–south comparison

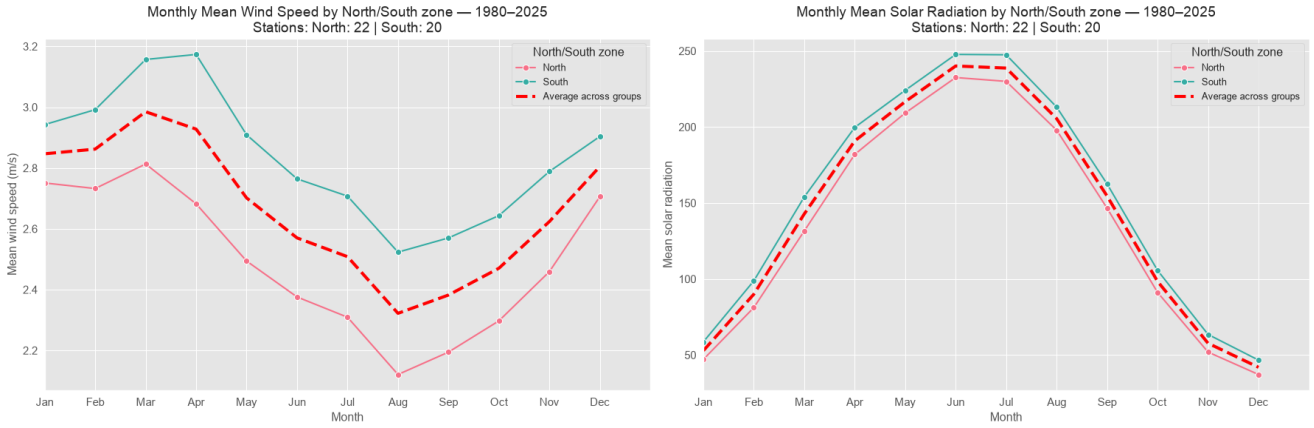


Figure 7: Monthly profiles by latitude group, 1980–2025.

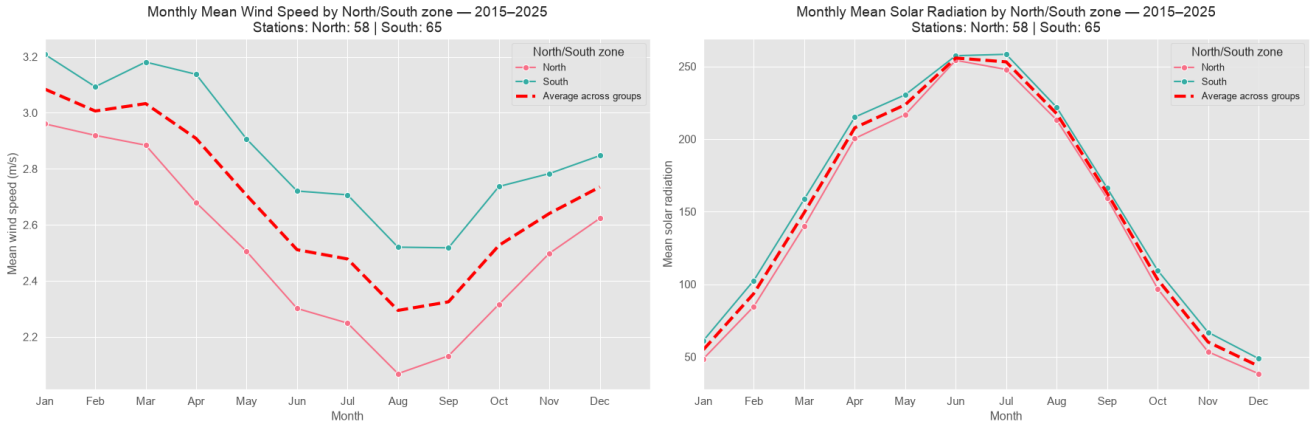


Figure 8: Monthly profiles by latitude group, 2015–2025.

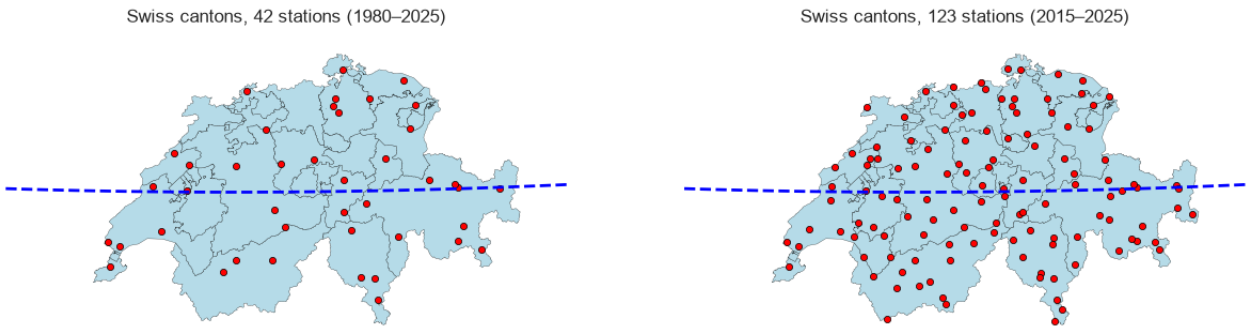


Figure 9: North–south stratification for 1980–2025 (left) and 2015–2025 (right).

2.4 Four geographic sectors

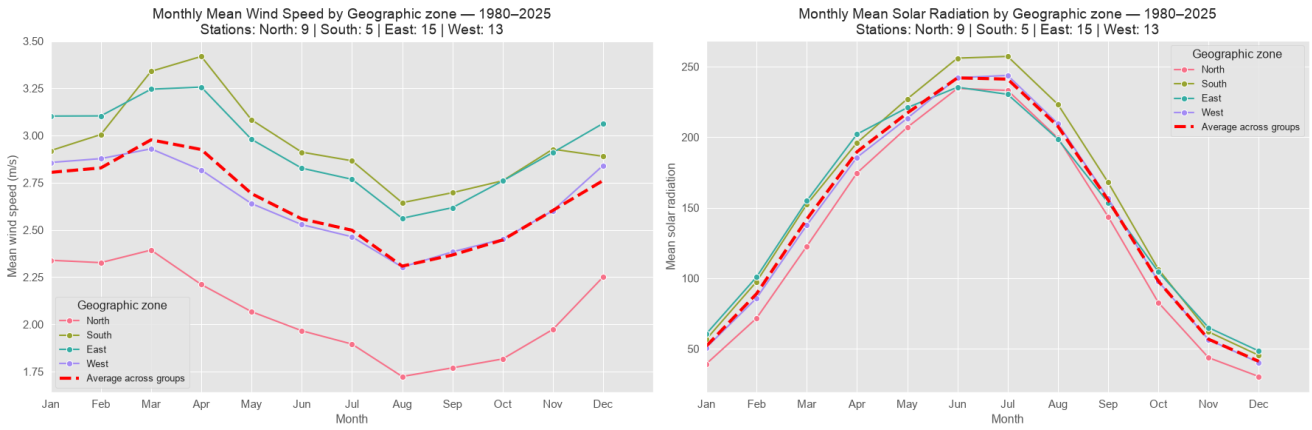


Figure 10: Monthly profiles by geographic sector, 1980–2025.

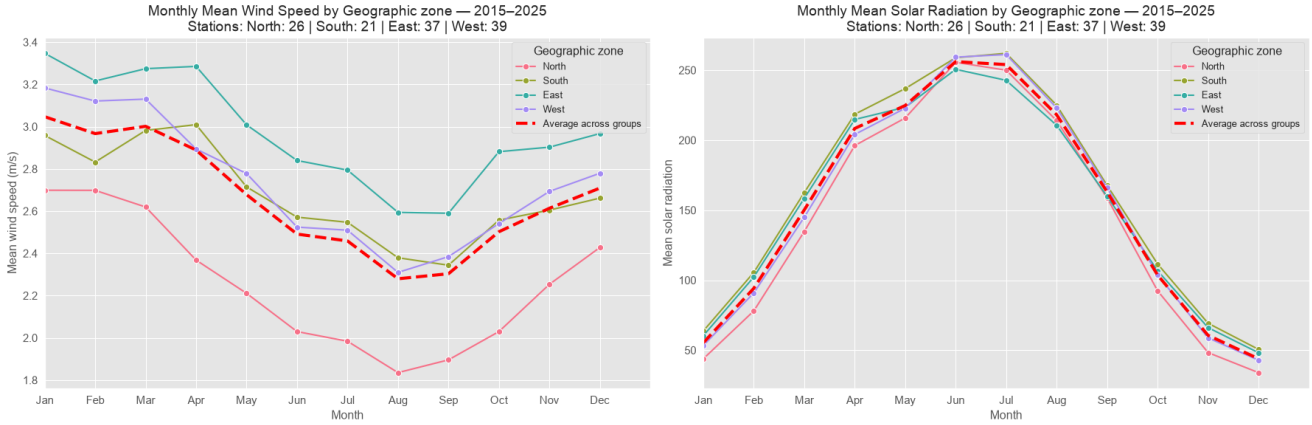


Figure 11: Monthly profiles by geographic sector, 2015–2025.

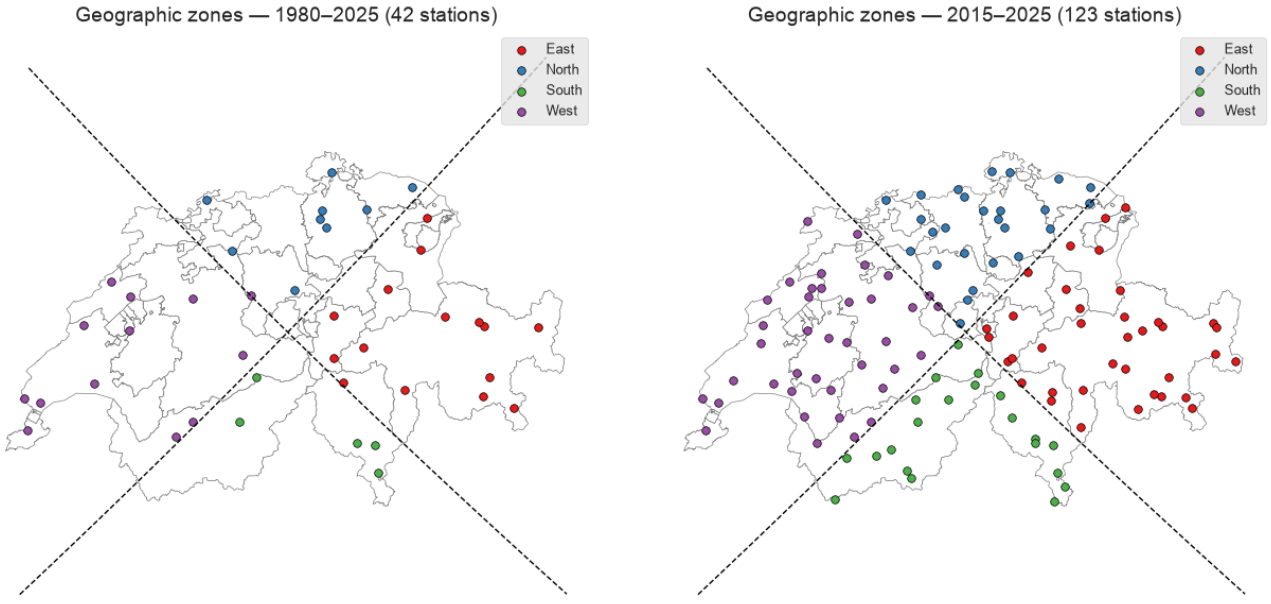


Figure 12: Four-sector stratification for 1980–2025 (left) and 2015–2025 (right).

A Station coverage

A.1 Network development

The bar charts below show an increase from 72 wind stations in 1980 to about 150 in 2025, and from 50 solar stations in 1981 to about 134 in 2025. Wind coverage rises sharply around 1993, while solar coverage expands strongly after 2008.

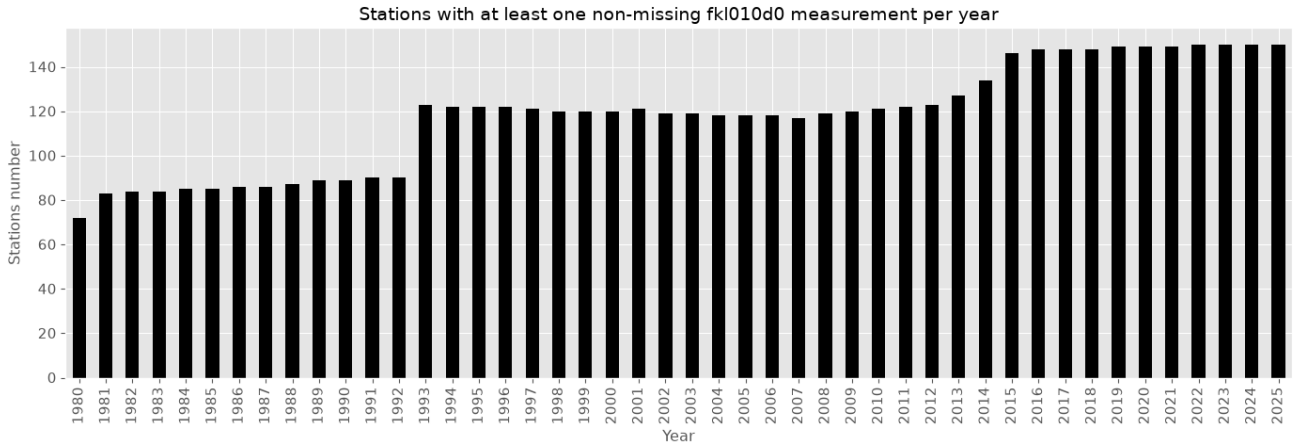


Figure 13: Stations with at least one daily wind measurement in each year.

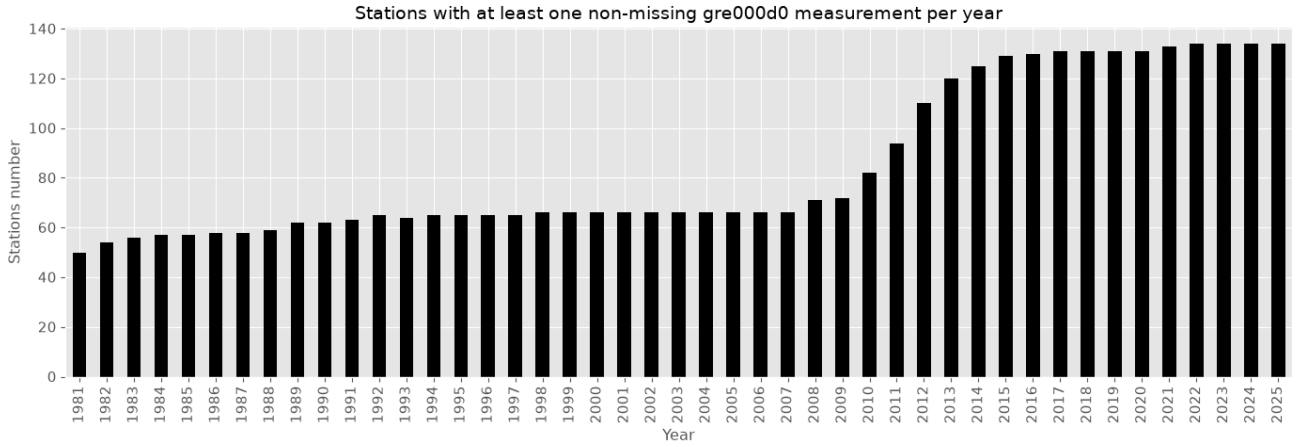


Figure 14: Stations with at least one daily solar-radiation measurement in each year.